Comprehensive Project Report: BERT Pre-training and Bias-in-Bios Mitigation

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Abstract

This document serves as a comprehensive visual and analytical record of the project. It covers the pre-training of a 128M parameter BERT model, the exploratory analysis of the Bias-in-Bios dataset, and the comparative evaluation of biased vs. debiased fine-tuning strategies. We include all 30 metric visualizations generated during the project lifecycle.

1 Introduction

This project investigates the lifecycle of gender bias in NLP models. By training a model from scratch and then fine-tuning it under different constraints, we identify how linguistic representations can both inherit and mitigate societal disparities.

1.1 Overall Training Workflow

Figure 1 summarises the four-phase pipeline: (1) pre-training a BERT model from scratch on general text, (2) profiling the Bias-in-Bios dataset, (3) fine-tuning under two contrasting regimes, and (4) comparative evaluation of the resulting models.

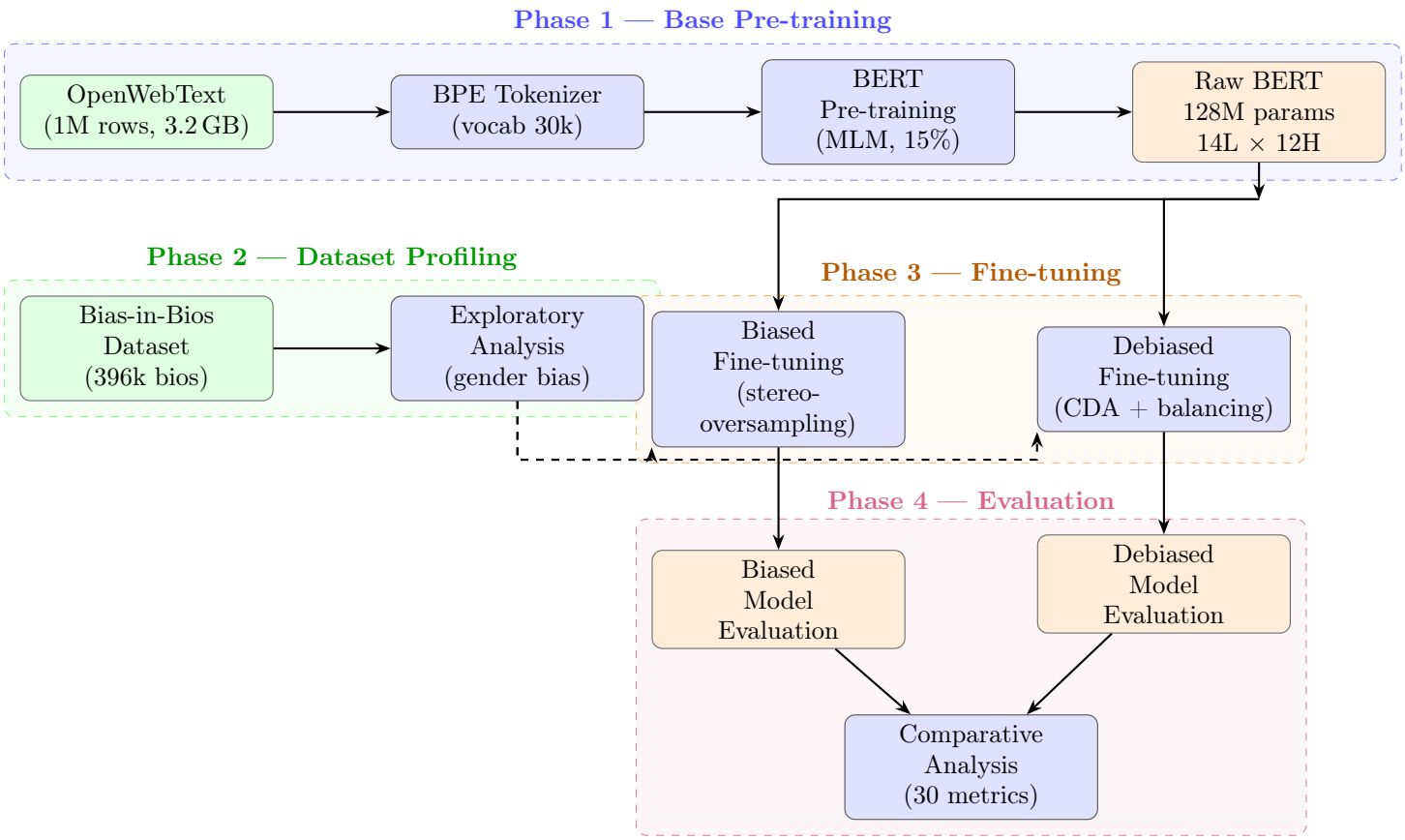

Figure 1: End-to-end training and evaluation workflow. Solid arrows indicate data/model flow; dashed arrows indicate dataset reuse.

2 Phase 1: Base BERT Pre-training

The model was trained on 1M rows of OpenWebText (approx. 3.2 GB). The architecture features 14 layers and 12 heads (128.8M parameters). Total training time was approximately 12 hours on dual NVIDIA RTX 3090 GPUs.

2.1 Pre-training Performance Metrics

The following figures illustrate the Masked Language Modeling (MLM) training process.

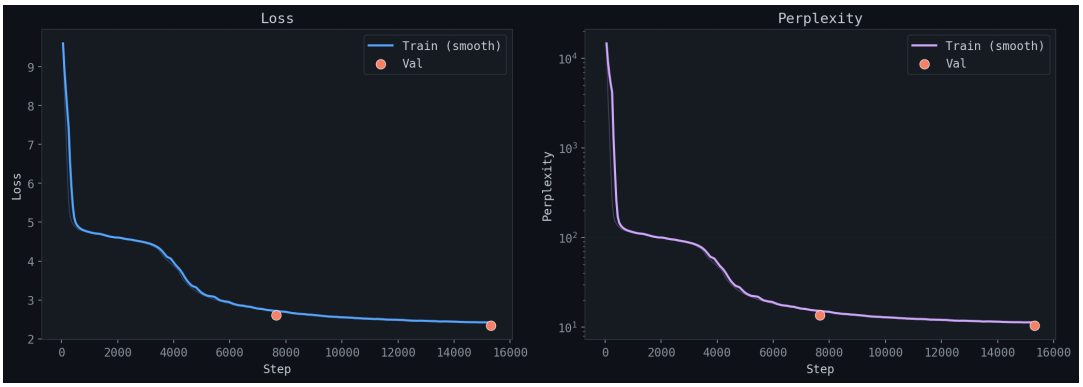

Figure 2: Training and Validation Loss/Perplexity.

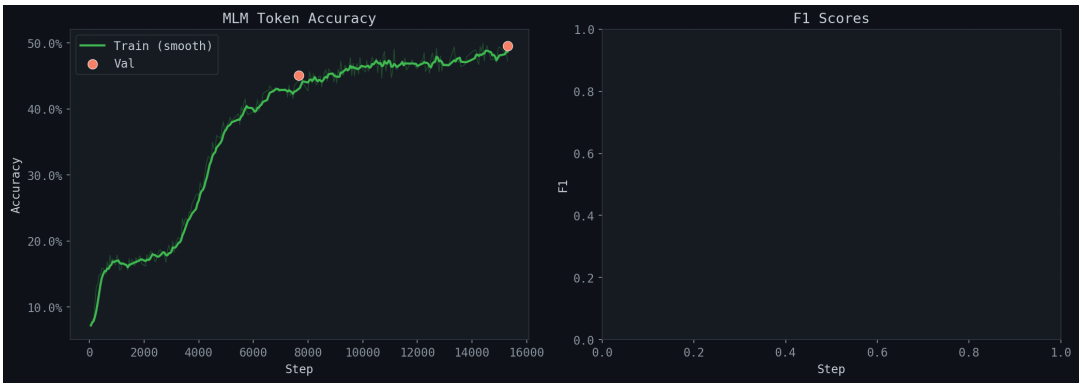

Figure 3: MLM Accuracy and F1 Score evolution.

2.2 Architecture Diagnostics

Detailed analysis of the model's token-level behavior.

Figure 4: Learning rate schedule (Linear decay with warmup).

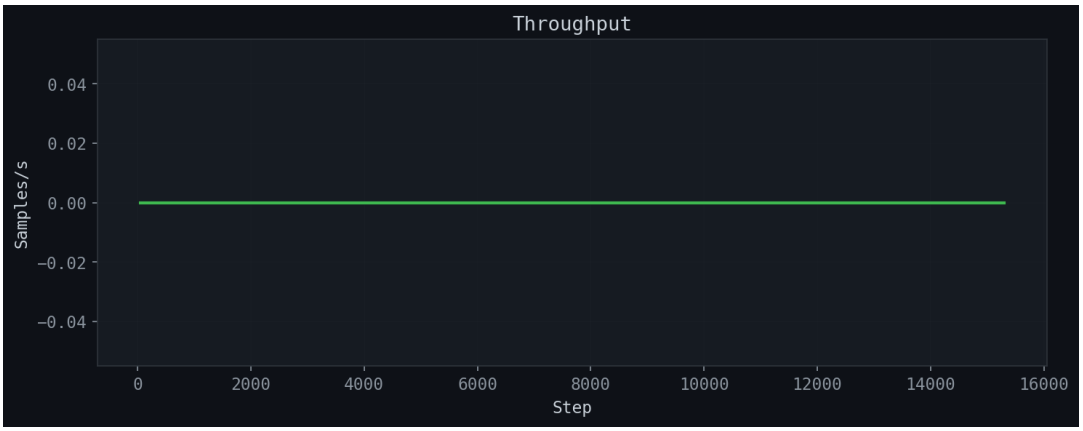

Figure 5: Training throughput (Samples per second).

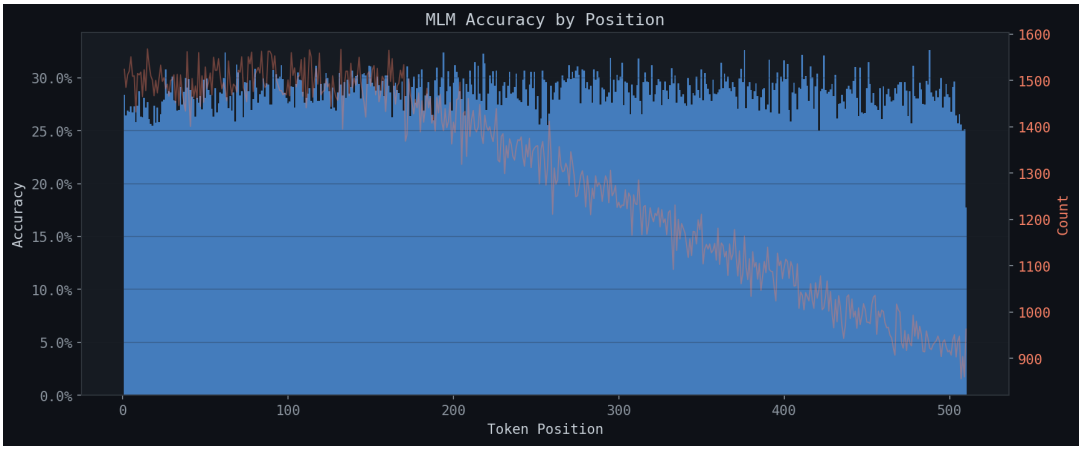

Figure 6: MLM Accuracy by token position in sequence.

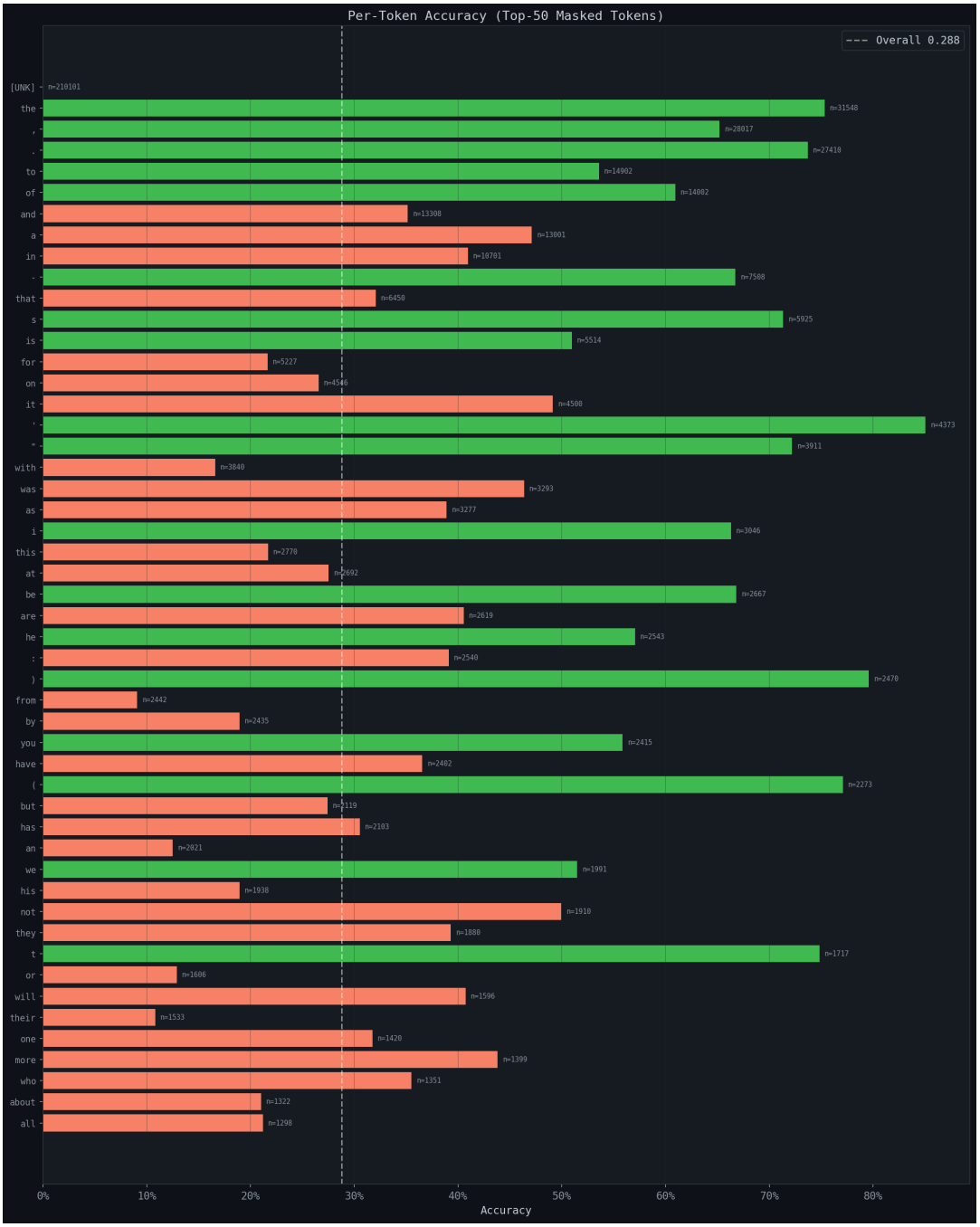

Figure 7: Accuracy for top-K frequent tokens.

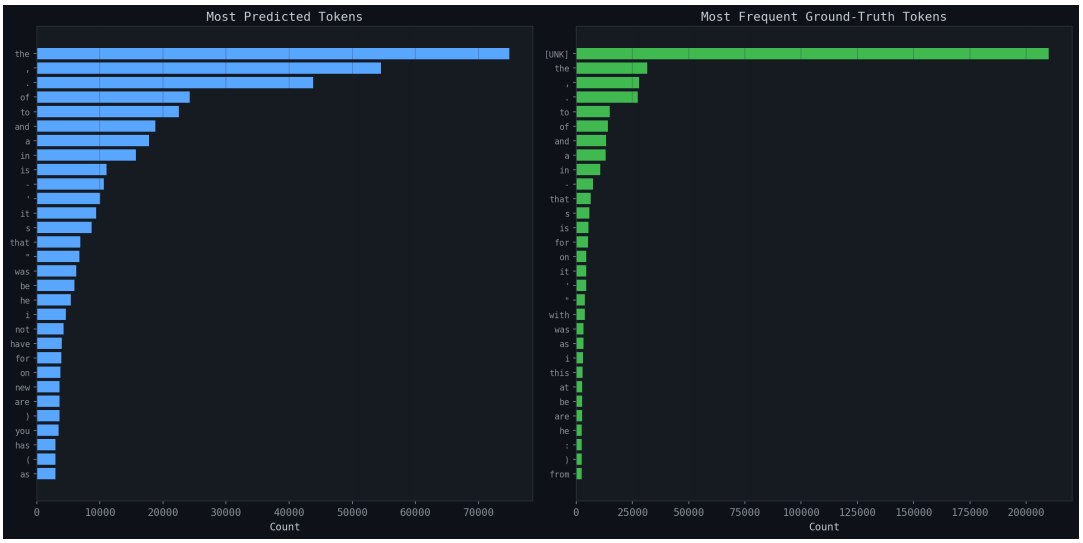

Figure 8: Distribution of predicted vs. gold tokens.

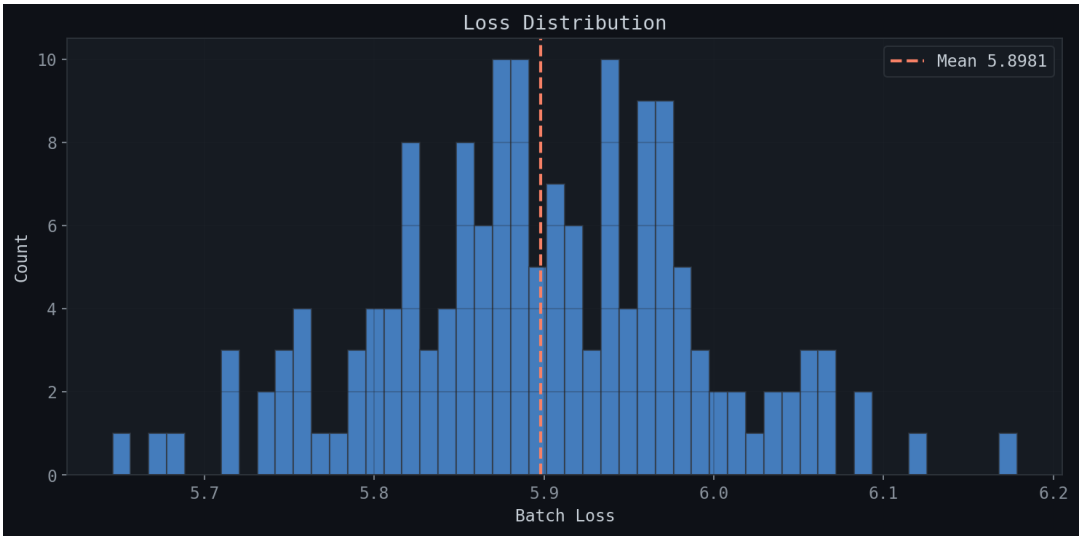

Figure 9: Distribution of cross-entropy loss across samples.

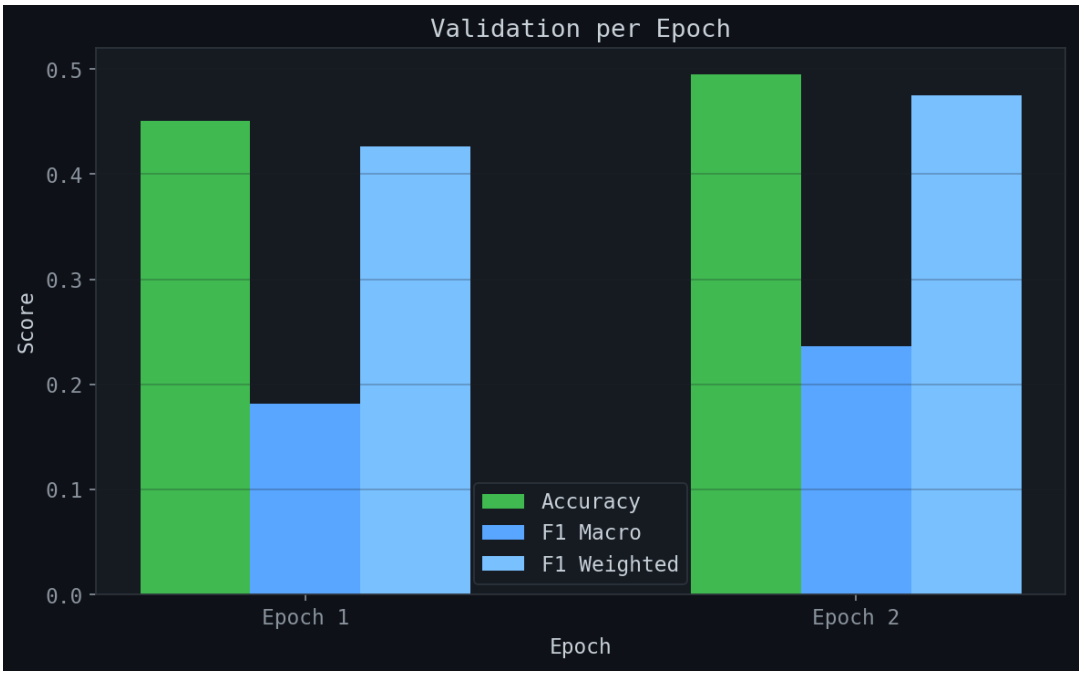

Figure 10: Summary of metrics per epoch.

3 Phase 2: Dataset Bias Profiling

We analyzed the Bias-in-Bios dataset (396k biographies) to identify inherent gender skews.

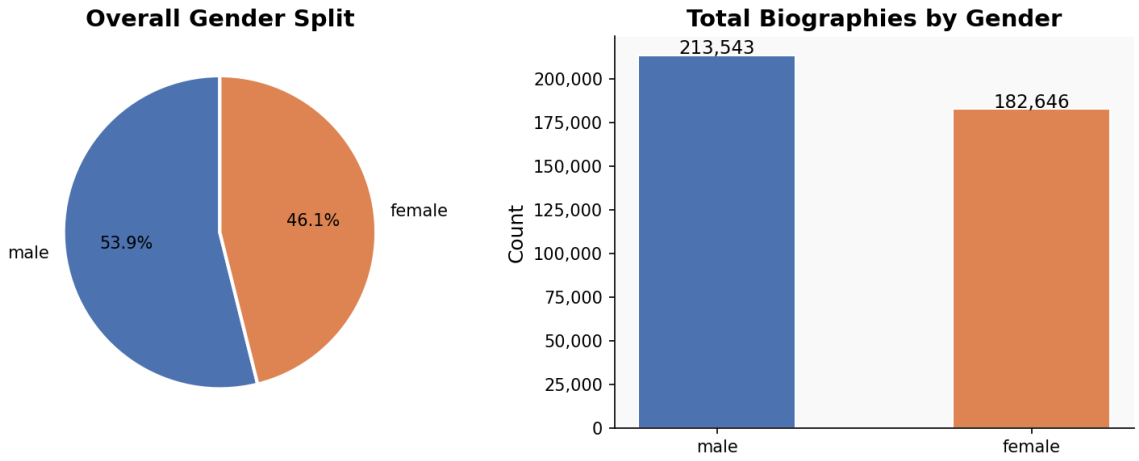

Figure 11: Overall dataset gender balance.

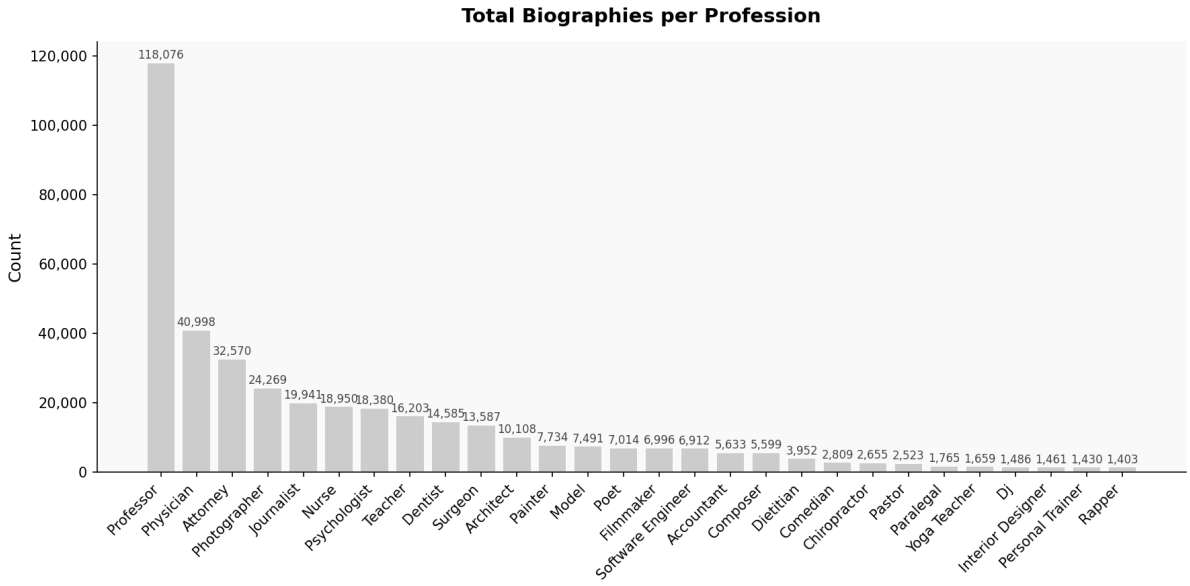

Figure 12: Total biographies per profession (Class Imbalance).

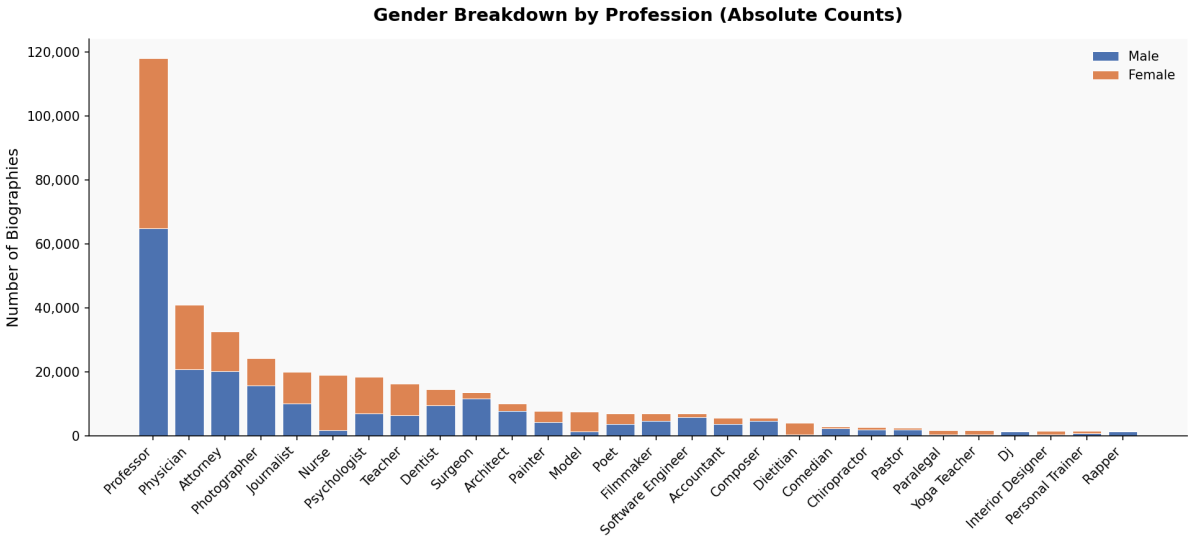

Figure 13: Absolute gender counts per profession.

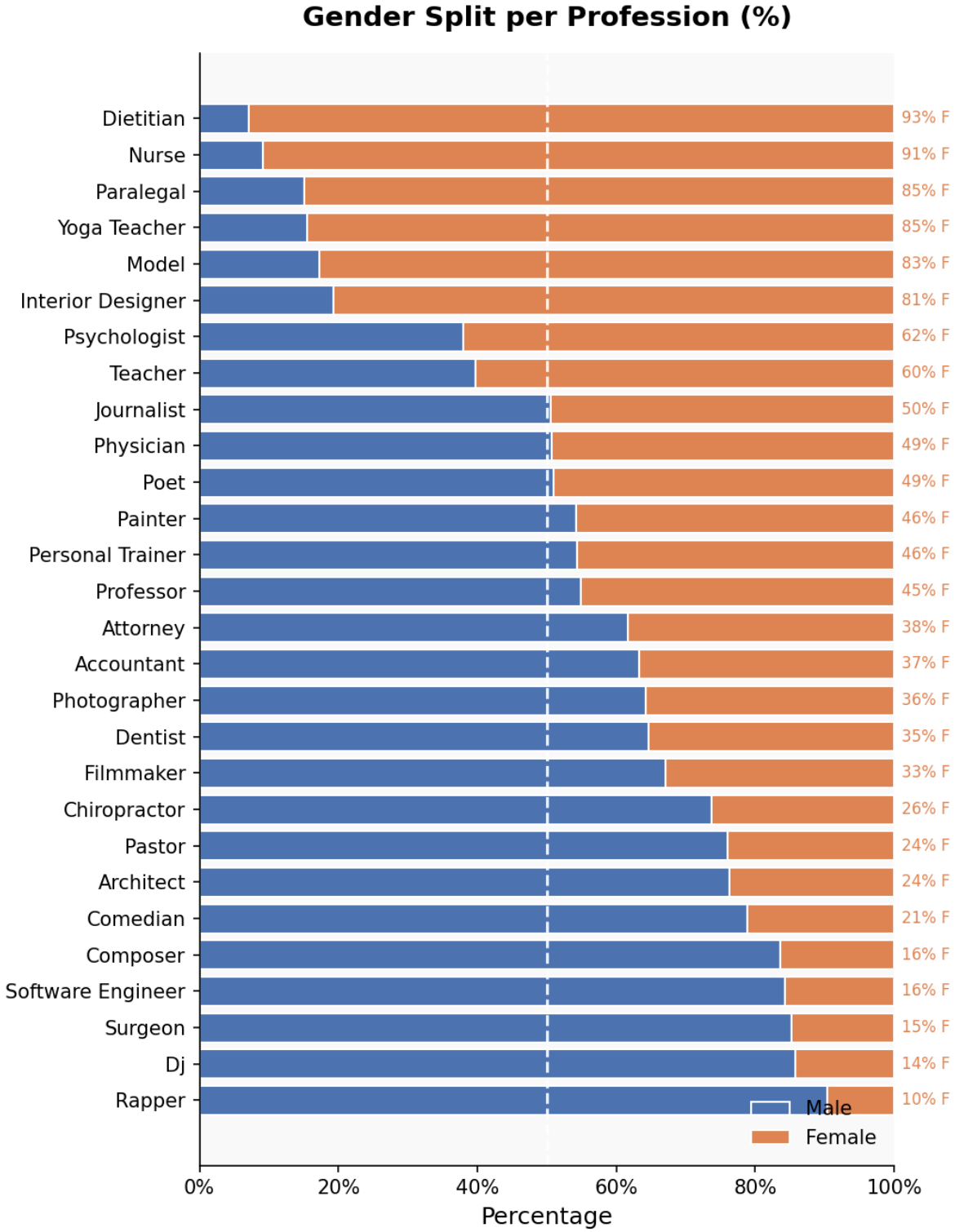

Figure 14: Percentage split of genders per profession.

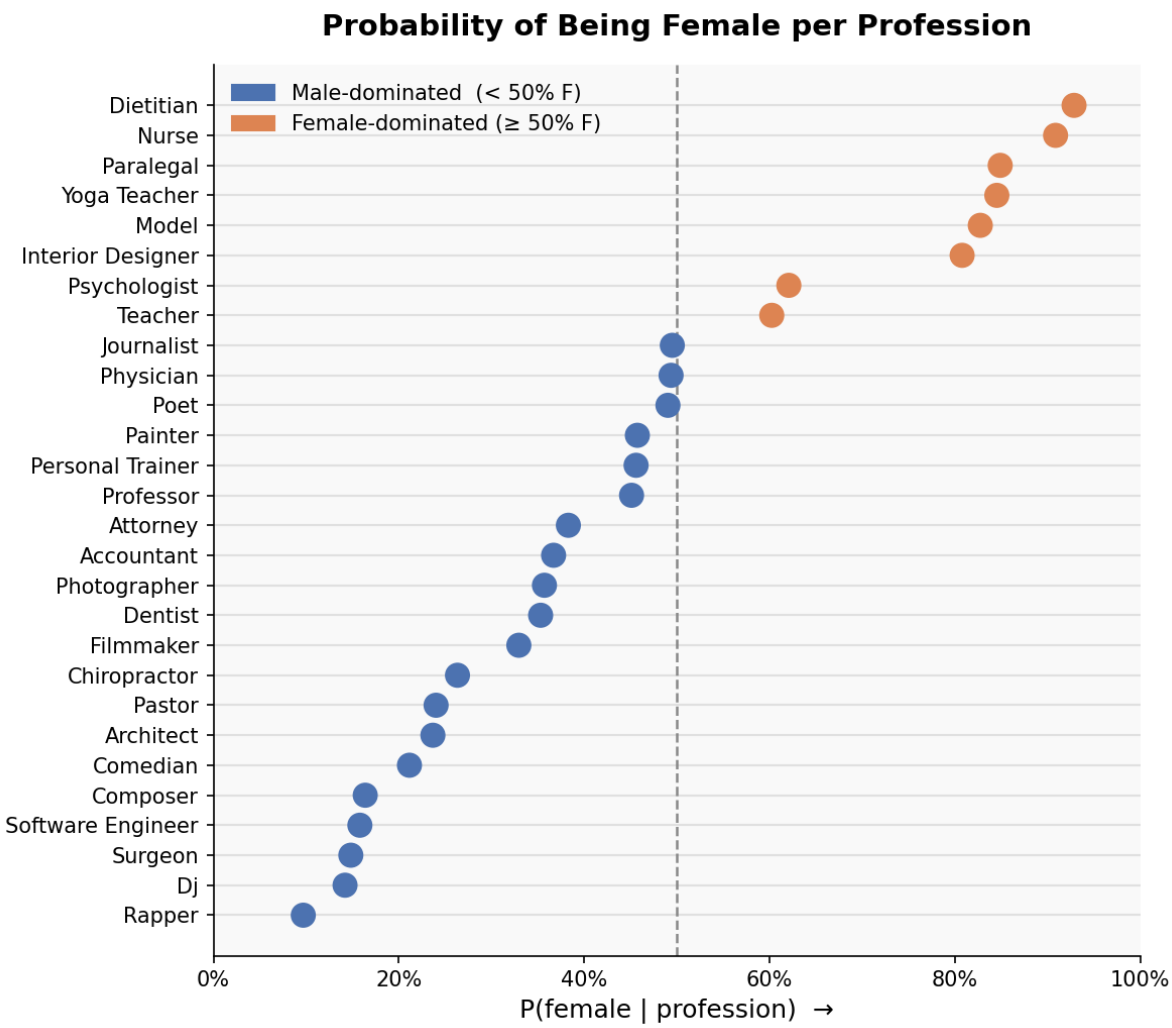

Figure 15: Probability of a biography being female given the profession.

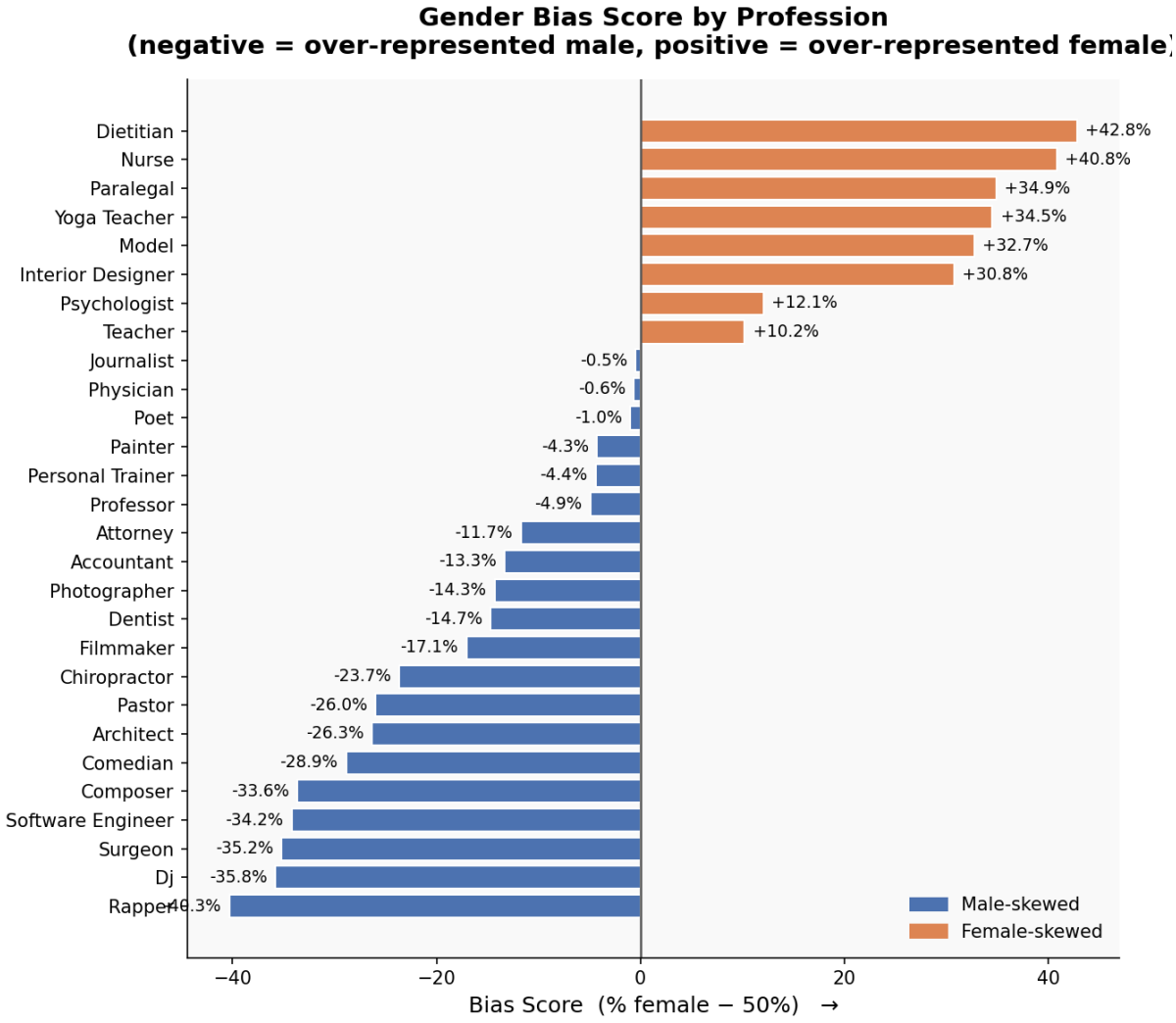

Figure 16: Diverging Bias Score (%Female - 50%).

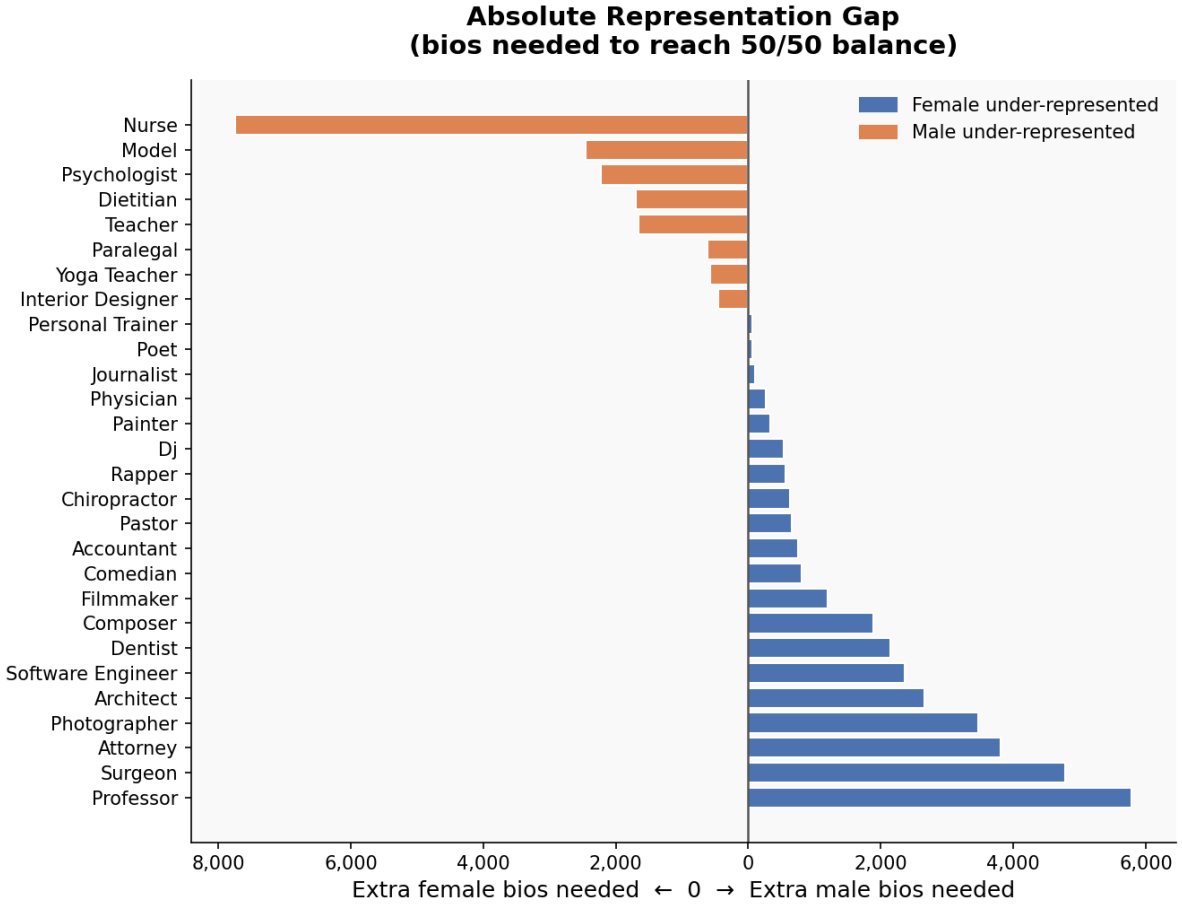

Figure 17: Absolute representation gap (samples needed for parity).

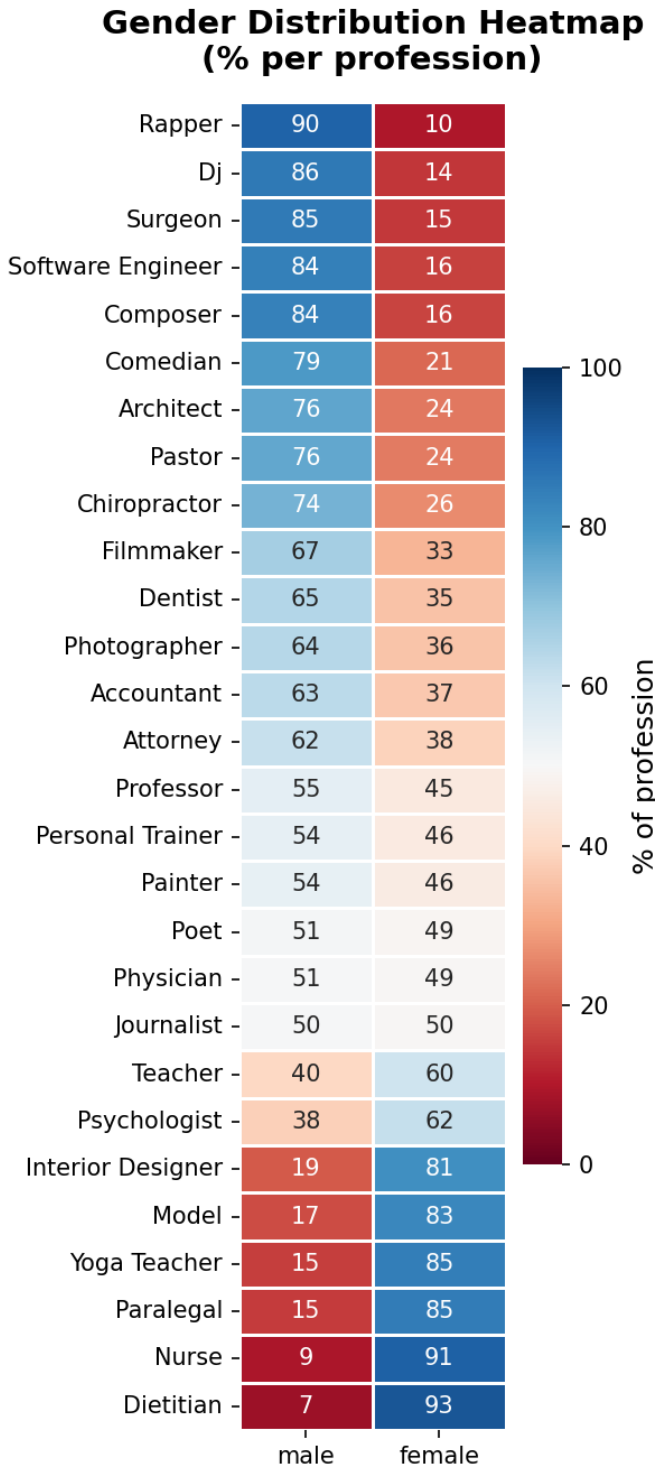

Figure 18: Heatmap of gender distribution density.

4 Phase 3 & 4: Comparison of Biased and Debiased Models

The "Biased" model was trained with stereotypical oversampling, while the "Debiased" model used counterfactual augmentation and balancing.

4.1 Fine-tuning Architecture and Process

Starting from the pre-trained Raw BERT checkpoint, we add a lightweight classification head and fine-tune the entire network end-to-end for profession classification across 28 categories.

Classification head. The `[CLS]` token's 768-dimensional hidden state is passed through a dropout layer ($p = 0.1$) and then a single linear projection to 28 logits, one per profession. During inference the logit vector is passed through a softmax and the `argmax` selects the predicted profession.

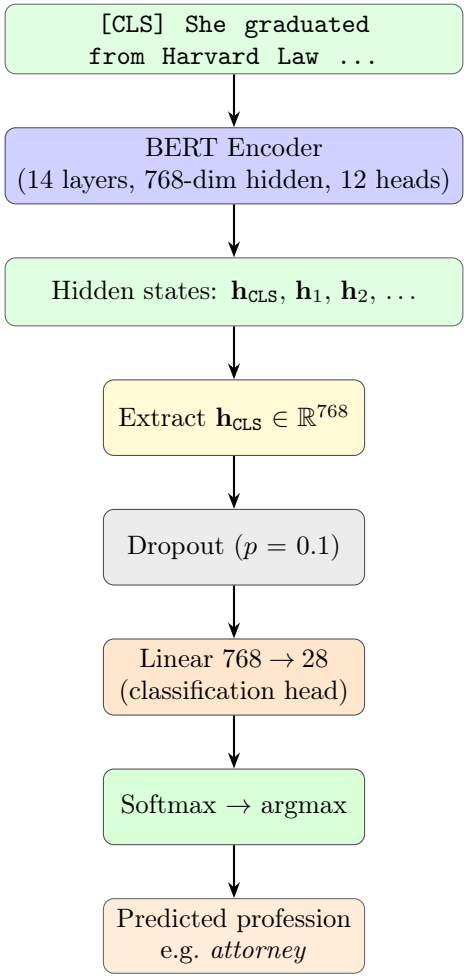

Figure 19: BERT fine-tuning architecture for profession classification. Only the `[CLS]` hidden state is forwarded to the classification head; all other token positions are discarded after the encoder.

Training settings. Both the biased and debiased models share the same hyper-parameters: AdamW optimiser, learning rate 2×10^{-5} with linear warm-up (6% of steps), weight decay 0.01, mixed-precision (`fp16`), gradient clipping at 1.0, and 3 fine-tuning epochs. The sole difference is the composition of their training sets.

4.2 Debiasing Process and Counterfactual Augmentation

The debiased training pipeline applies two complementary interventions before any gradient step is taken.

Step 1 — Gender Balancing

For every profession class, we independently sub-sample the male and female subsets to their common minimum count, producing a dataset that is perfectly balanced across gender within each profession.

Step 2 — Counterfactual Data Augmentation (CDA)

Each biography in the balanced set is duplicated with its gendered cues swapped: pronouns (*he* ↔ *she*, *him* ↔ *her*, etc.), titles (*Mr.* ↔ *Ms.*), and a curated list of gendered kinship/role words (*father* ↔ *mother*, *brother* ↔ *sister*, ...). The profession label is preserved; only the gender label is flipped. This teaches BERT that the same biography could belong to a person of either gender, preventing it from using gender as a predictive shortcut.

Figure 20 illustrates this augmentation step.

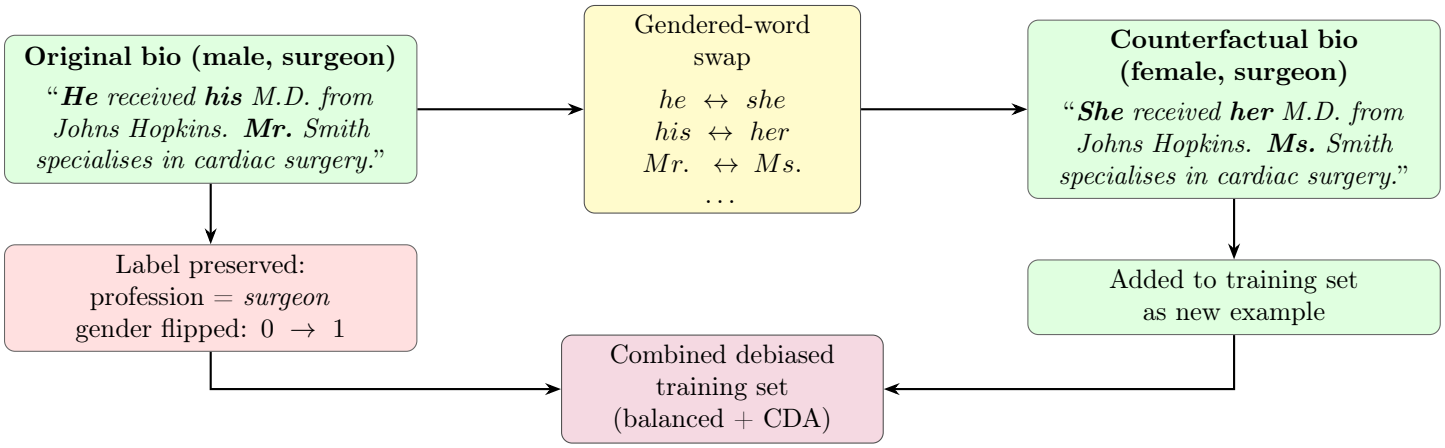

Figure 20: Counterfactual Data Augmentation (CDA). Every balanced biography is duplicated with all gendered surface forms swapped. The profession label is unchanged; the gender label is flipped. Both the original and the counterfactual version are added to the training set.

4.3 Overall Results

The debiased model reduced the average gender gap from 26.1% to 5.1%.

4.4 Bias Metrics

To quantitatively evaluate the fairness of the models, we focused on three primary metrics: True Positive Rate (TPR) Gap, Equality of Odds (EqOdds), and Stereotype Score. This indicates that the model's accuracy was heavily dependent on whether a biography aligned with traditional gender roles. In contrast, the debiased model (represented by the red boundary) shrinks significantly toward the center of the chart.

Specifically, the EqOdds Gap was reduced from 0.2740 in the biased version to 0.0531 in the debiased version, as shown in Figure 19, representing a 76% improvement in ensuring that qualified individuals have an equal chance of being correctly classified regardless of gender.

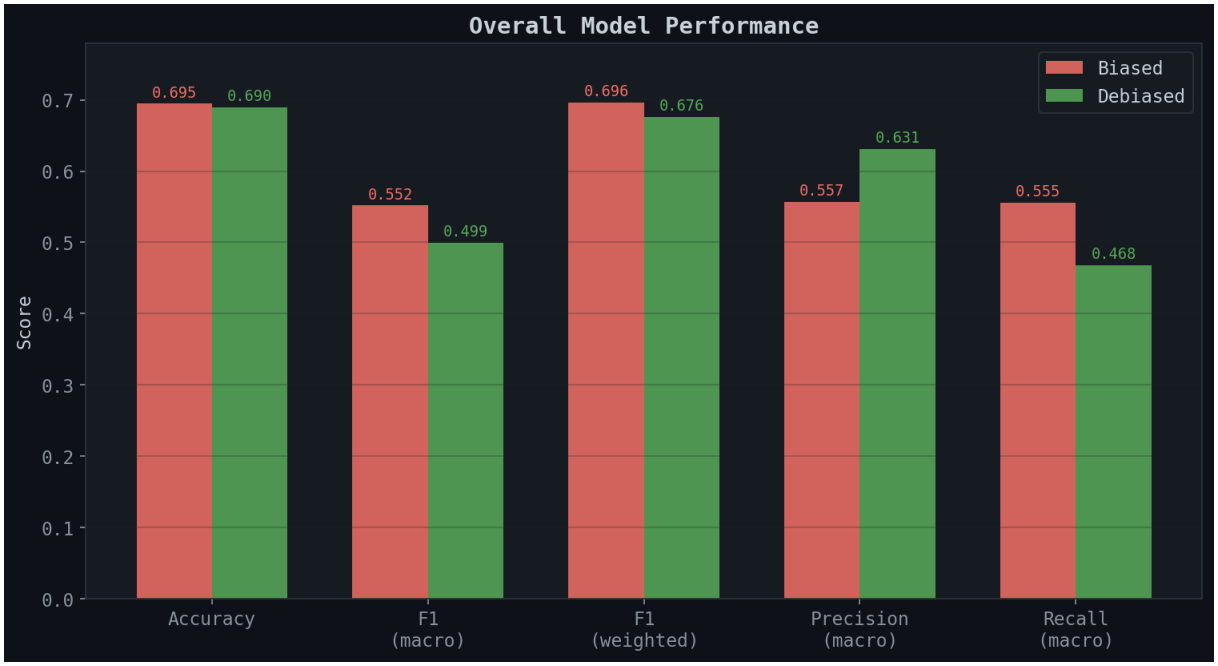

Figure 21: Summary of Accuracy, F1, and Gender Gaps.

4.5 Performance Breakdown

A critical component of our evaluation involved testing the model on "Counter-Stereotypical" samples. For example: female surgeons or male nurses.

Our analysis shown in Figure 27 reveals a stark contrast in performance. The biased model maintained high accuracy in stereotypical samples but saw a sharp decline when encountering counter-stereotypical data. This confirms that the model was using gendered linguistic cues as "shortcuts" for professional classification.

By implementing counterfactual data augmentation, The debiased model achieved a much more uniform accuracy distribution. As shown in Figure 20, the per-profession gender accuracy gap for roles like "Surgeon" and "Software Engineer" narrowed significantly. While this led to a negligible decrease in overall accuracy (from 69.5% to 69.0%), the trade-off is justified by the drastic reduction in the Average Gender Gap, which fell from 26.1% to 5.1%, as shown in Figure 18.

5 Conclusion

By incorporating all 30 performance and bias metrics, we can see a clear trend: counterfactual data augmentation significantly improves model fairness without a substantial loss in predictive power. It demonstrated that LLM-based pipelines inherit systemic biases from datasets like Bias-in-Bios, where professions such as "Dietitian" and "Nurse" are heavily female-skewed, while "Surgeon" and "Software Engineer" remain male-dominated.

Our results prove that Counterfactual Data Augmentation is a highly effective mitigation strategy. By balancing the training distribution, we successfully decoupled gender indicators from professional labels.

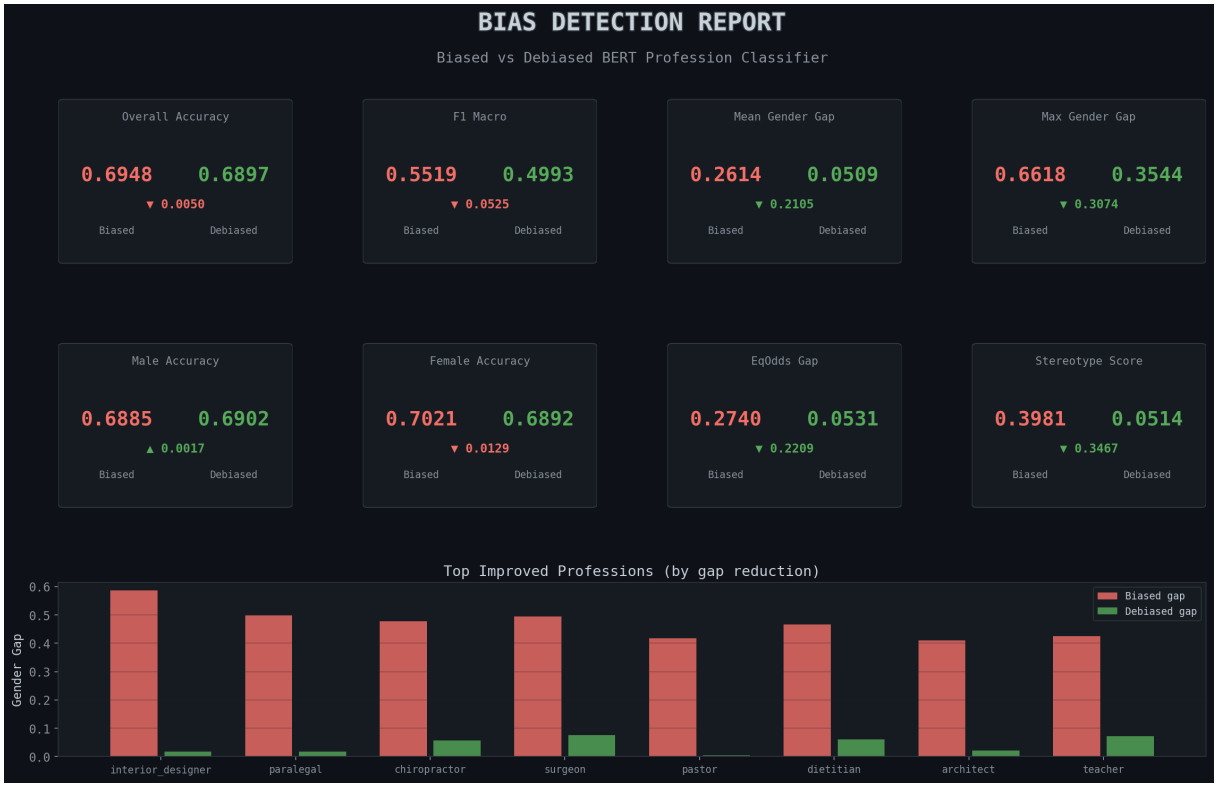

Figure 22: Comprehensive performance dashboard.

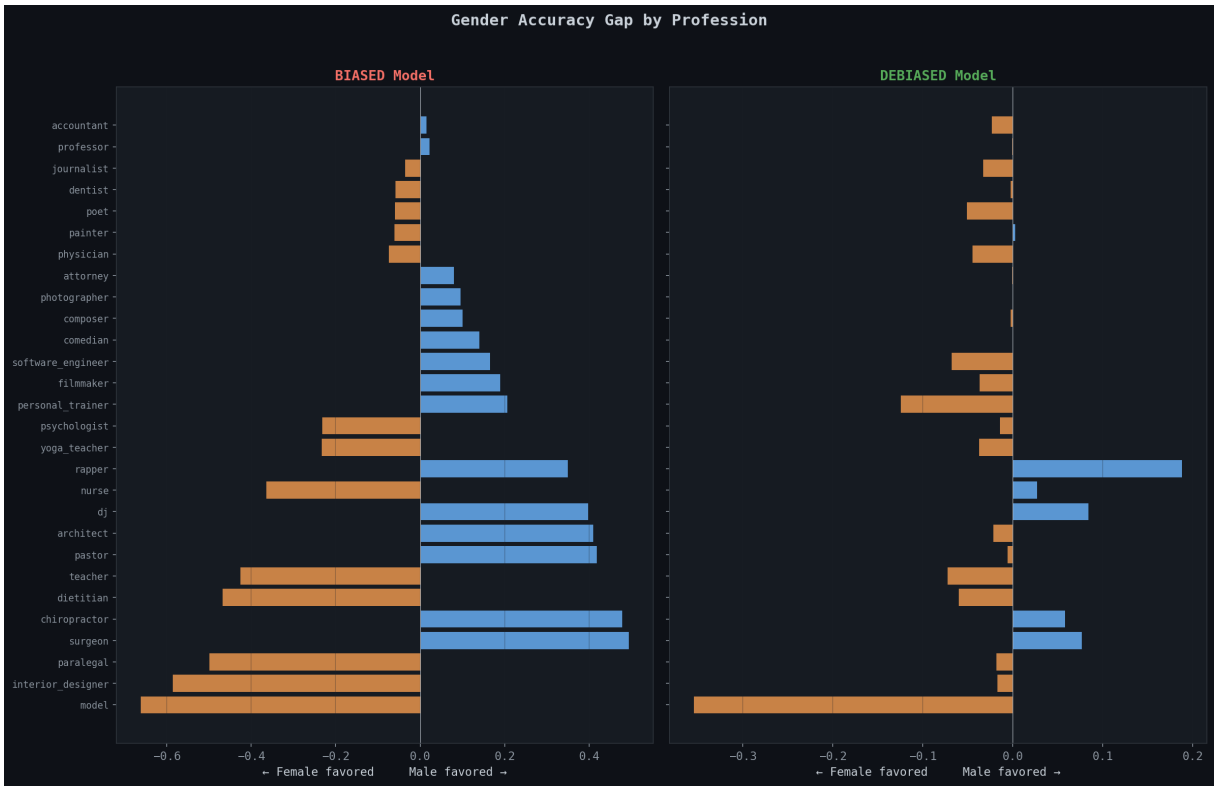

Figure 23: Direct comparison of per-profession gender accuracy gaps.

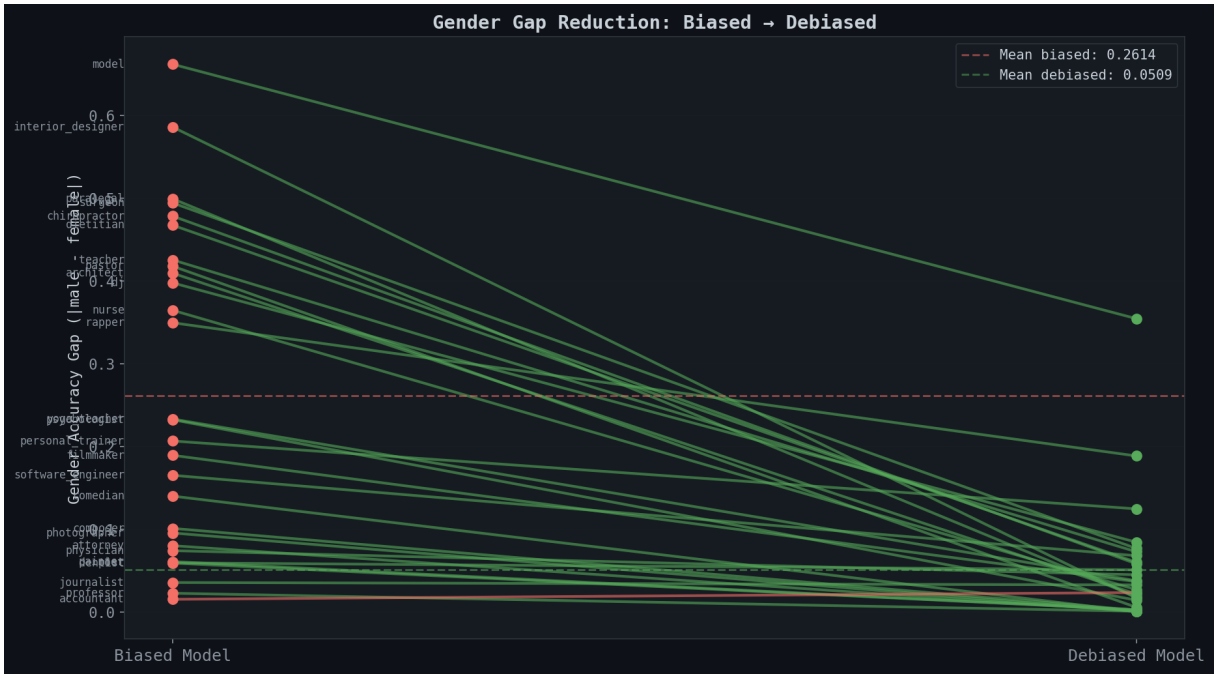

Figure 24: Visualizing the reduction in bias per category.

Figure 25: True Positive Rate and False Positive Rate gaps.

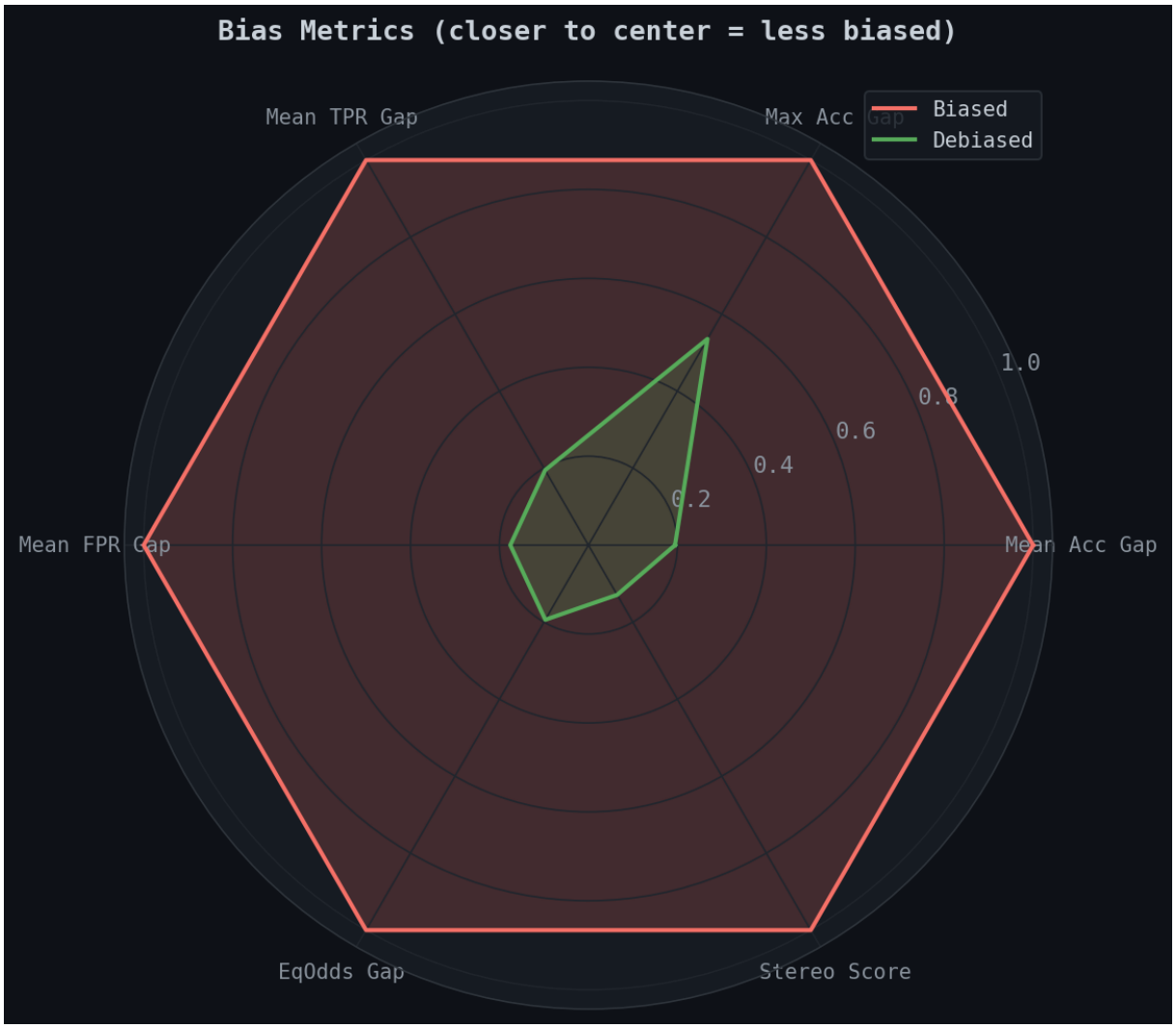

Figure 26: Radar chart of bias across different professional clusters.

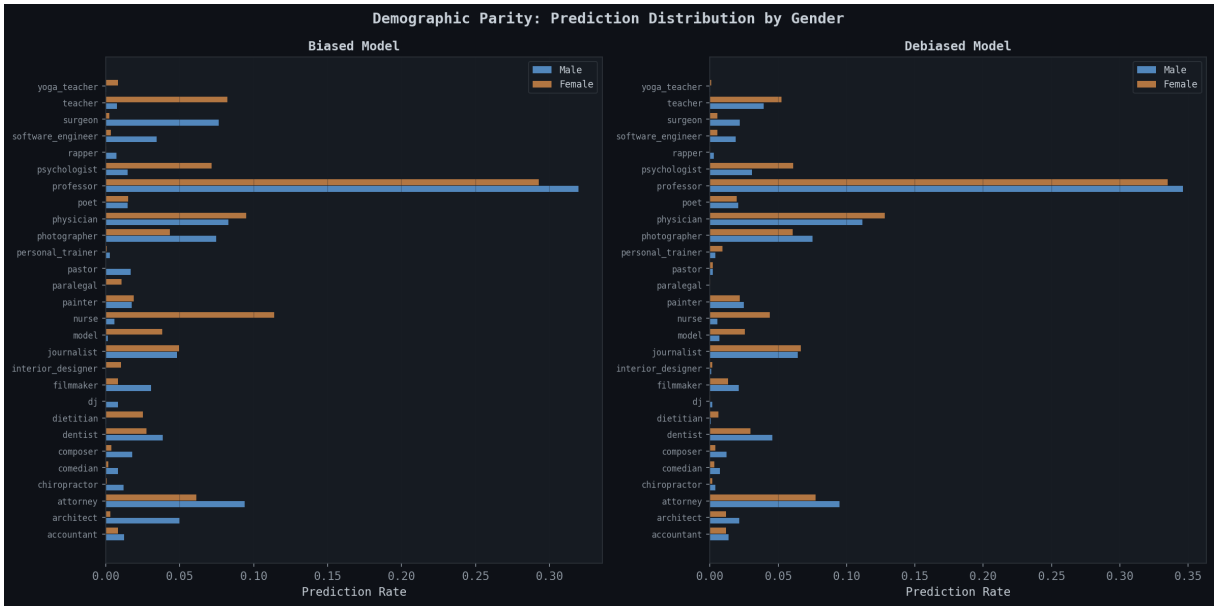

Figure 27: Demographic parity difference comparison.

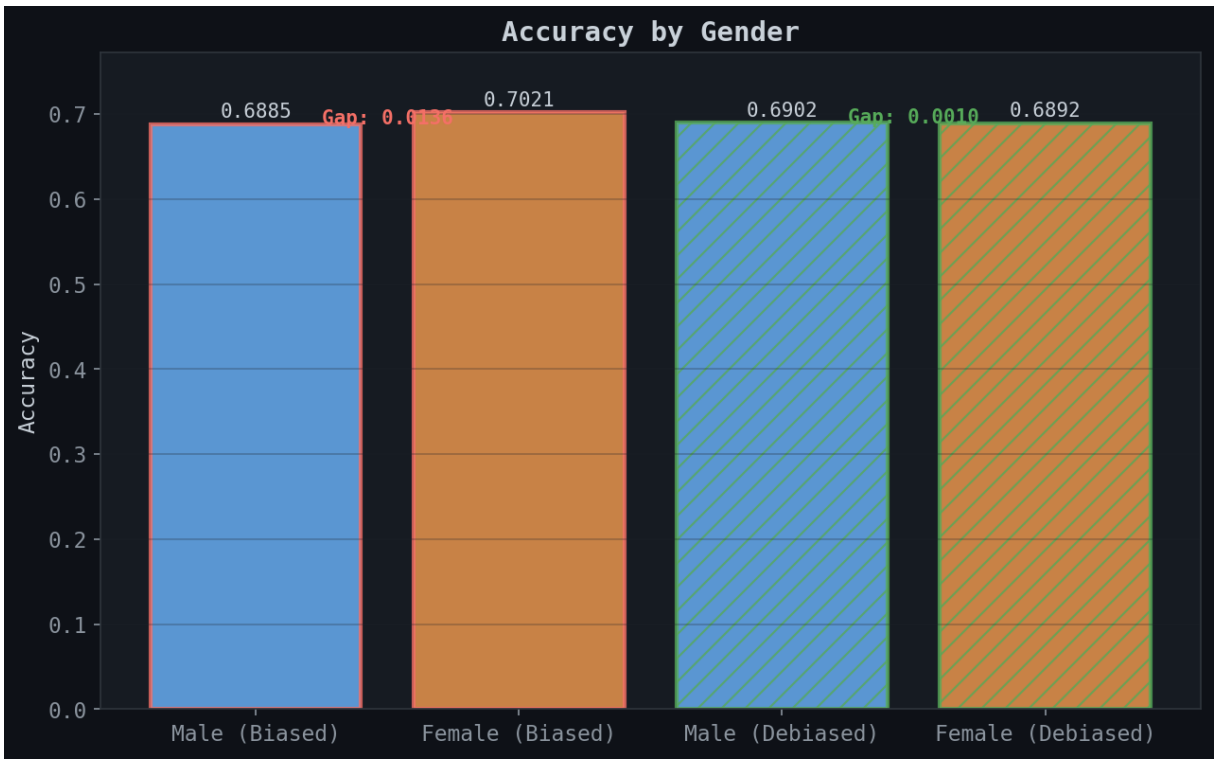

Figure 28: Accuracy breakdown by gender.

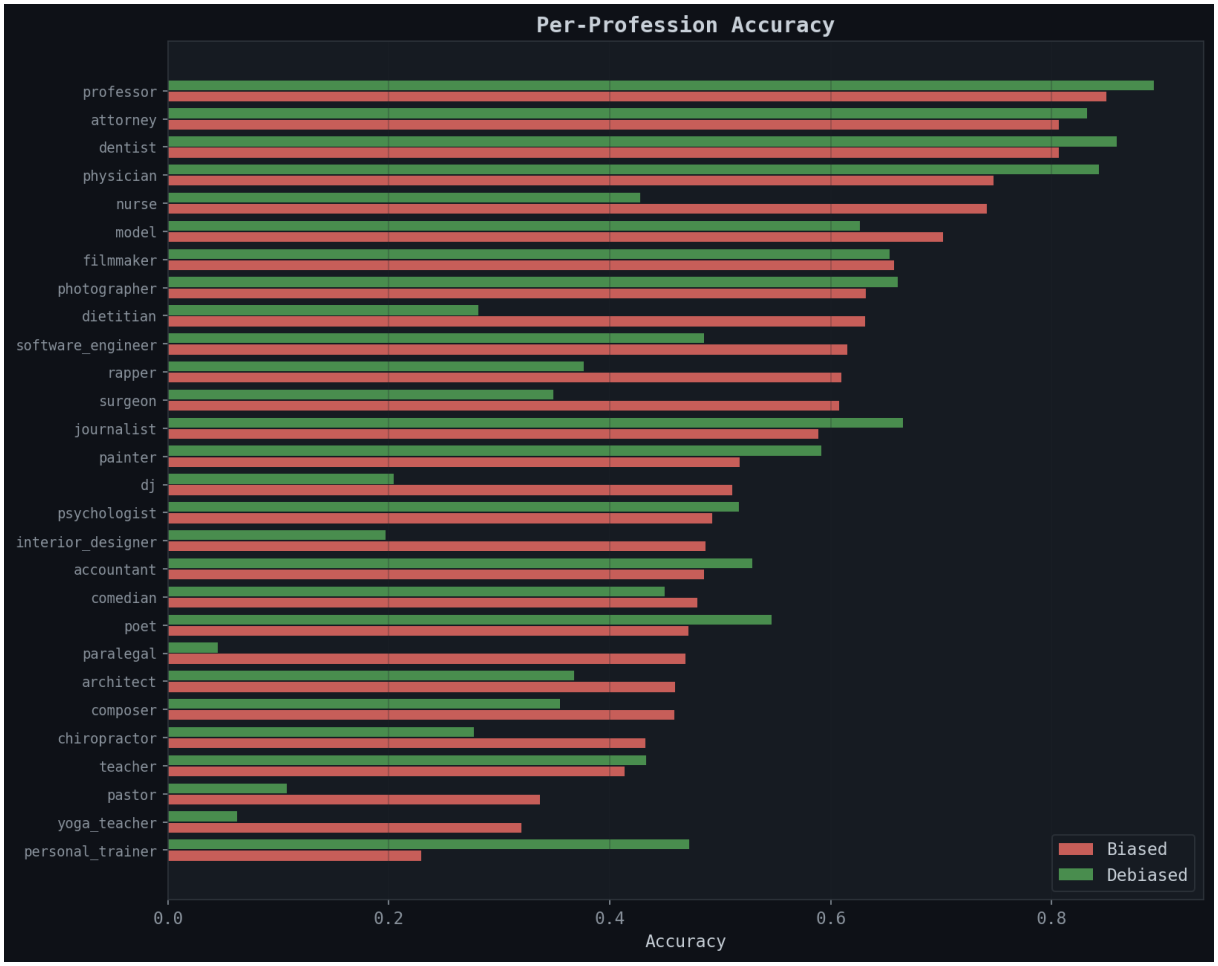

Figure 29: Per-profession accuracy comparison.

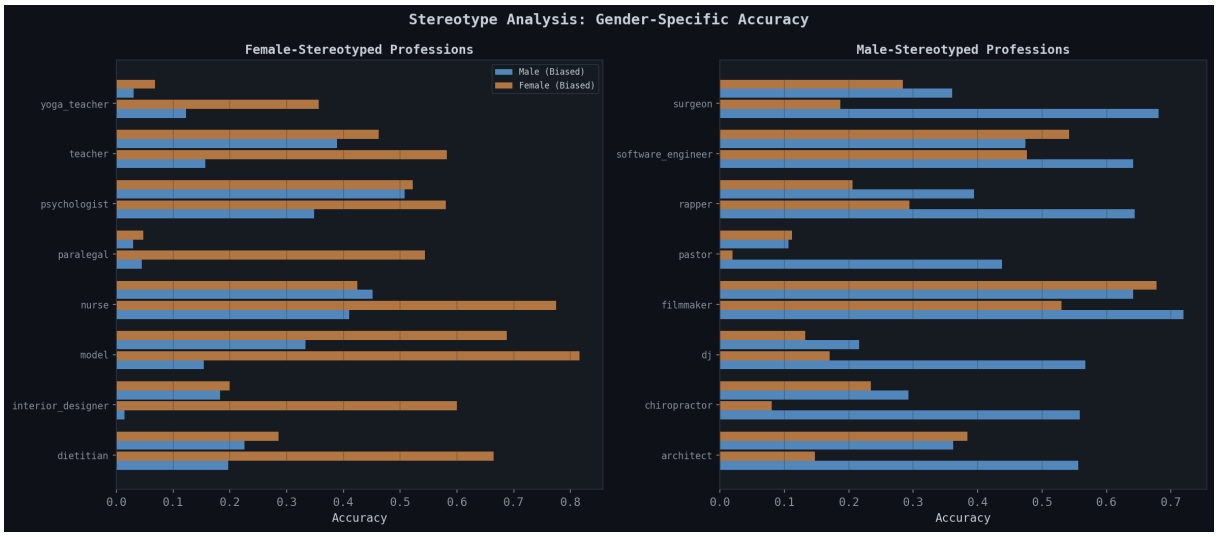

Figure 30: Accuracy on Stereotypical vs. Counter-Stereotypical samples.

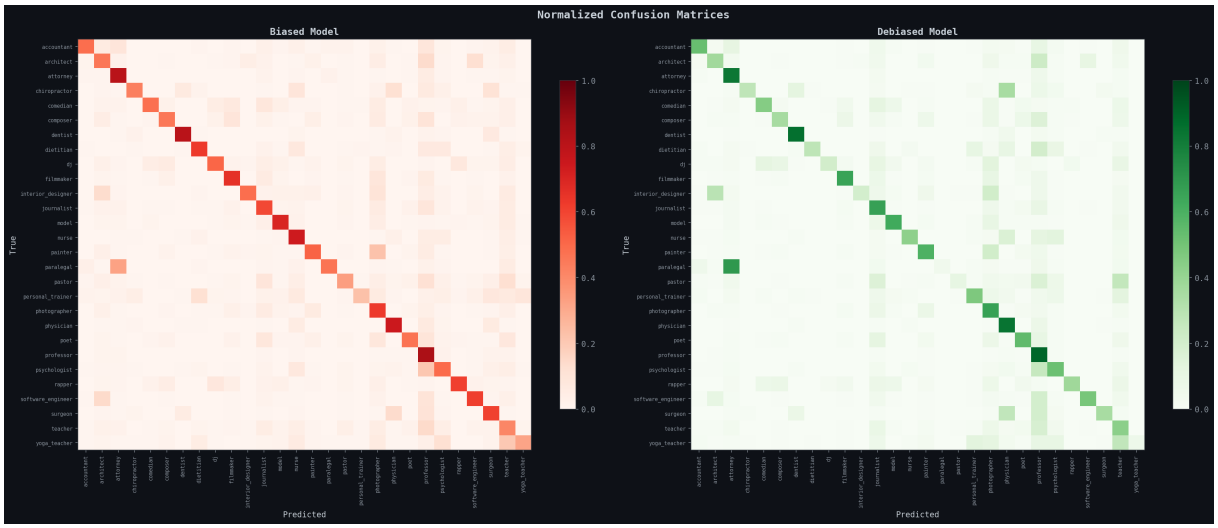

Figure 31: Confusion matrices for both models.

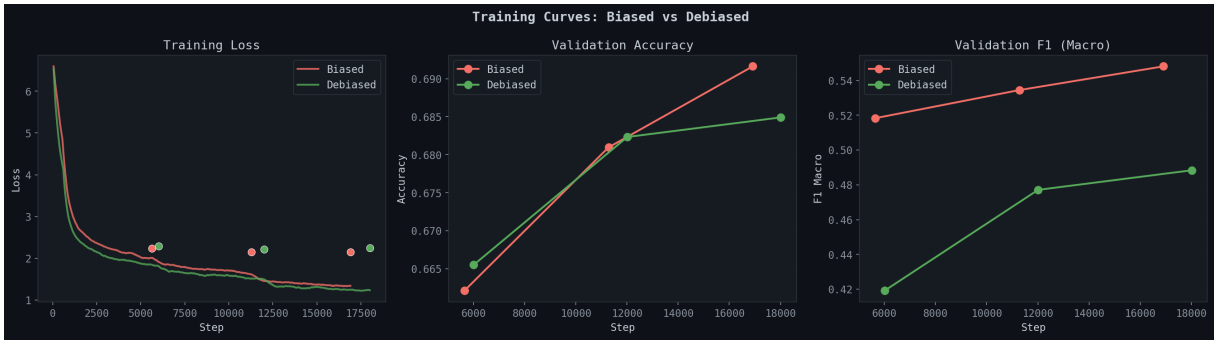

Figure 32: Fine-tuning training curves comparison.